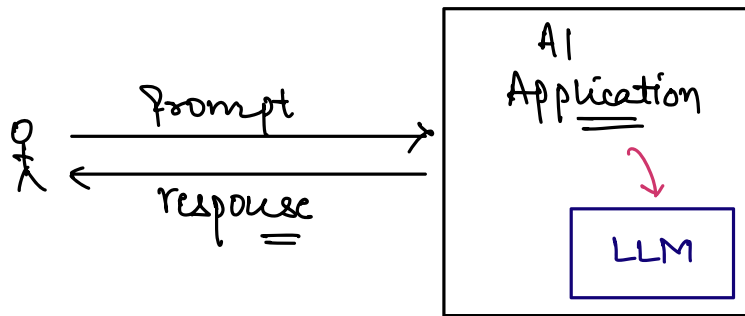


# # Agentic Systems



Query: Please find the cheapest flight from DEL to BLR for next week, book it for me if it is in my budget (~6000 INR) & add it to my calendar.

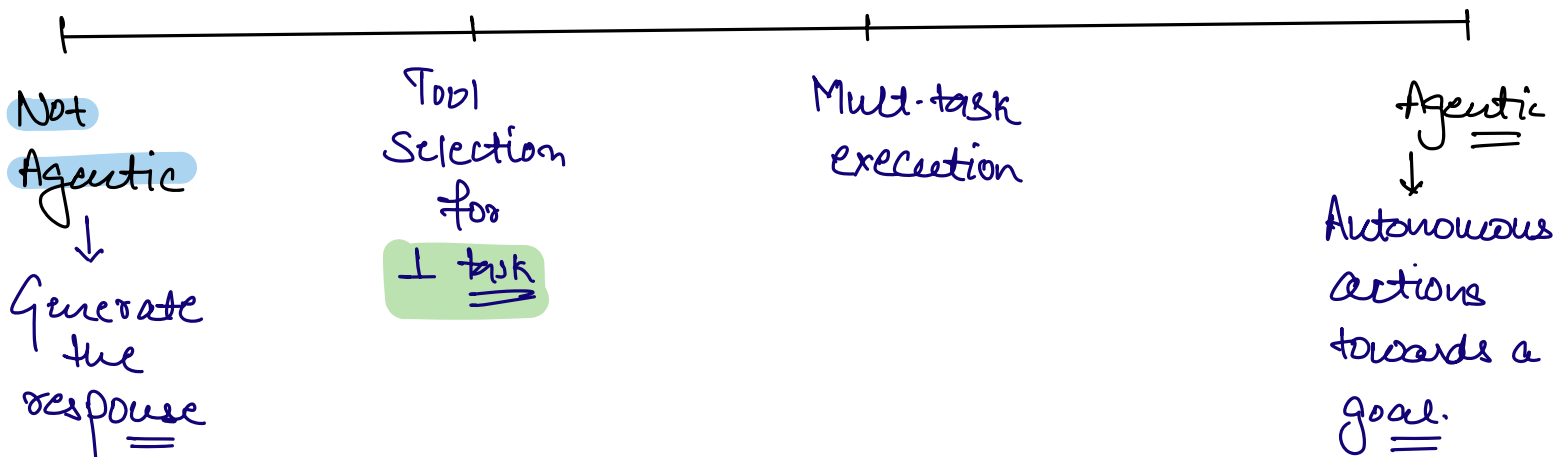
## Agentic System

↳ There are agents which can take autonomous decisions.

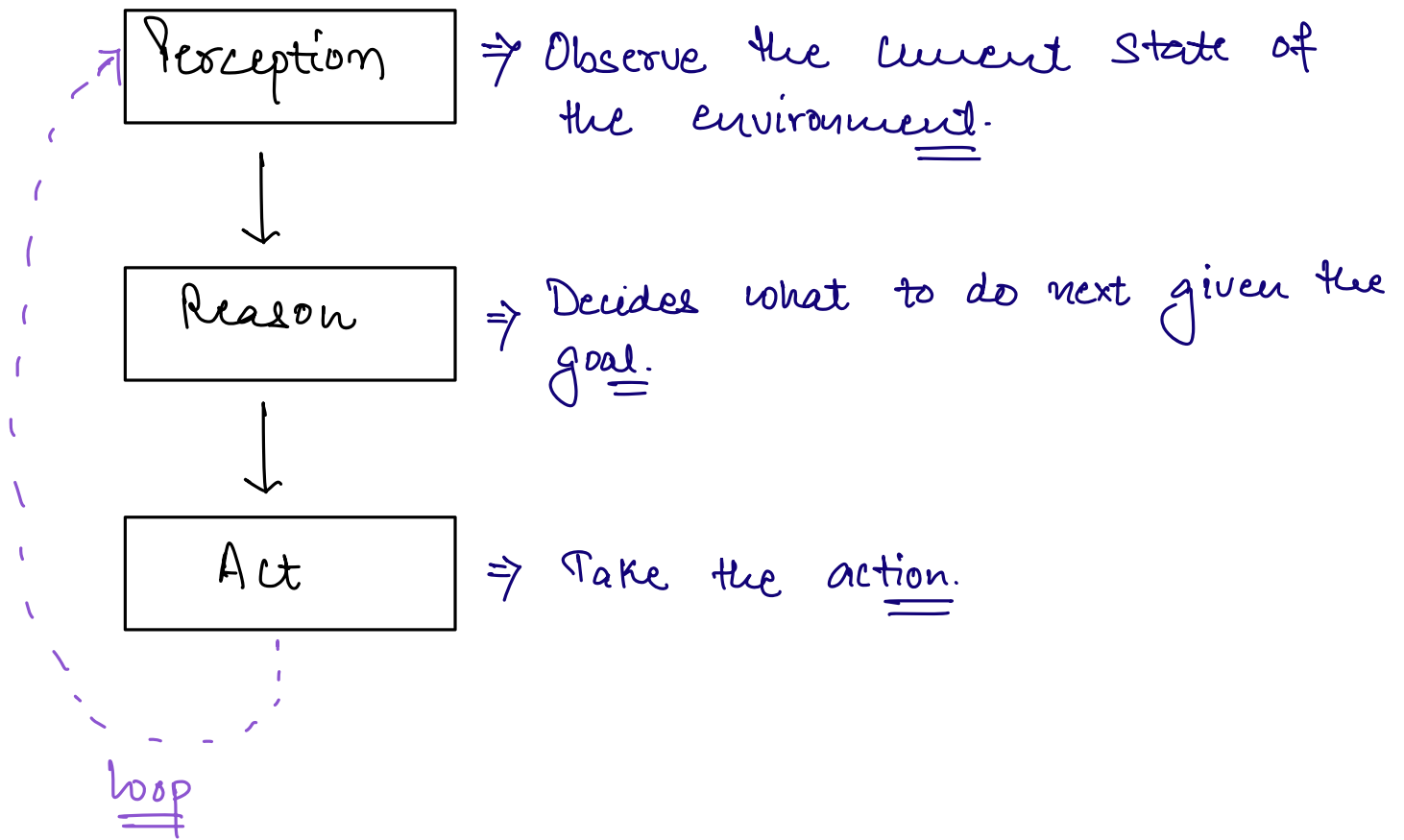
↓  
independent.

Agentic system is a system built around LLM that can autonomously take a series of actions towards a goal rather than just generating the text.

→ System decides on its own about the decision that needs to be taken.



# # Perception - Reasoning - Action loop.



Perceive: Gathering information about the current State.

Reason: LLM processes this information against the goal & decides what to do next.

Act: Doing something that changes the state.

Goal: "Find the cheapest flight to Tokyo next month, book it if under \$800, and add it to my calendar."

Cycle 1:

PERCEIVE: initial goal, no information yet

REASON: "I need to search for flights first"

ACT: call flight\_search\_tool(destination="Tokyo", month="next")

Cycle 2:

PERCEIVE: flight search results – 3 options, cheapest is \$650

REASON: "\$650 is under the \$800 budget, I should book it"

ACT: call book\_flight\_tool(flight\_id="XJ204")

Cycle 3:

PERCEIVE: booking confirmation received, confirmation number ABC123

REASON: "Booking succeeded, now I need to add this to the calendar"

ACT: call add\_calendar\_event\_tool(event="Flight to Tokyo", date=...)

Cycle 4:

PERCEIVE: calendar event created successfully

REASON: "All parts of the goal are complete"

ACT: report completion to the user, stop looping

## # Planning & Reasoning



Decides what to do next.

Agent generates  
a sequence of  
intended steps  
towards a goal before  
executing them.

Without explicit planning:

Reason about step 1 → Act → Reason about step 2 (from scratch) → Act → ...

Risk: the agent may take a locally sensible action that is globally inefficient or contradicts a later step

With explicit planning:

Reason: "Here is my full plan: 1) search flights, 2) check budget, 3) book if valid, 4) add to calendar"

Then: Act on step 1 → check against the plan → Act on step 2 → ...

Benefit: each action is chosen in the context of the whole goal, not just the immediately preceding observation

- Planning means generating a sequence of intended steps toward a goal before executing them, rather than deciding purely one step at a time
- Planning happens within the reasoning step of the loop, typically at the start of a task and revisited as new information arrives
- A plan is a starting hypothesis, not a guarantee — a well-designed agent revises its plan when observed outcomes contradict its assumptions
- Planning and ongoing reasoning work together: planning alone becomes brittle, reasoning alone becomes shortsighted

# Memory.

→ Stateless

→ Stateful

⇒ By default, LLMs are stateless.

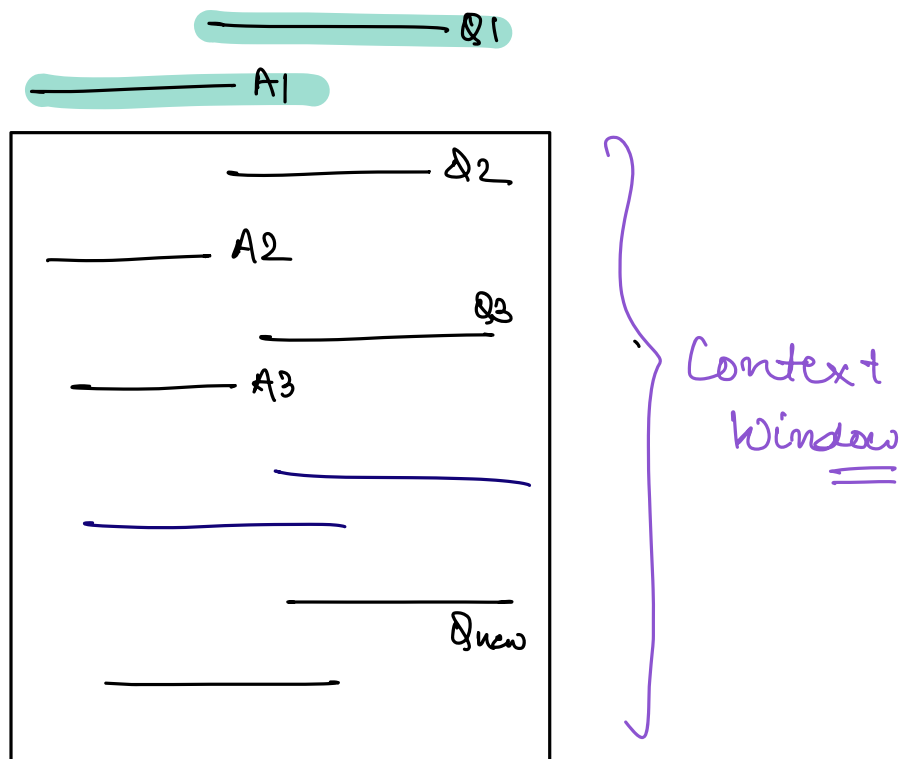
## # Types of Memory

→ Short-Term Memory

- Current Conversations.
- Exists only for the duration of the current task or session.
- Conversational Memory.

→ Long-Term Memory.

- Stored outside the model & retrieved when required.
- Survives across conversations or sessions.



- Short-term memory is the conversation's context window — bounded, and does not survive past the current session
- Long-term memory is stored persistently outside the model and retrieved only when relevant, often via embeddings and similarity search
- Long-term memory retrieval directly reuses the embedding and similarity pattern from the embeddings session, now applied to an agent's own past experience
- Retrieving only relevant memories, rather than including everything, follows the same context-optimization discipline covered in the cost optimization session